In [36]:
```python
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as pyplt
```

In [37]:
```python
df = sns.load_dataset('titanic')
```

In [38]:
```python
df.head()
```

Out[38]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_m
0	0	3	male	22.0	1	0	7.2500	S	Third	man	T
1	1	1	female	38.0	1	0	71.2833	C	First	woman	Fa
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	Fa
3	1	1	female	35.0	1	0	53.1000	S	First	woman	Fa
4	0	3	male	35.0	0	0	8.0500	S	Third	man	T

In [39]:
```python
df.describe()
```

Out[39]:

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

In [40]:
```python
df.isnull().sum()
```

Out[40]:
```
survived         0
pclass           0
sex              0
age            177
sibsp            0
parch            0
fare             0
embarked         2
class            0
who              0
adult_male       0
deck           688
embark_town      2
alive            0
alone            0
dtype: int64
```

In [41]:
```
df['age']=df['age'].fillna(df['age'].mean())
```

In [42]:
```
df['embarked']=df['embarked'].fillna(df['embarked'].mode()[0])
```

In [43]:
```
df['embark_town']=df['embark_town'].fillna(df['embark_town'].mode()[0])
```

In [44]:
```
df =df.drop('deck',axis=1)
```

In [45]:
```
df.isnull().sum()
```

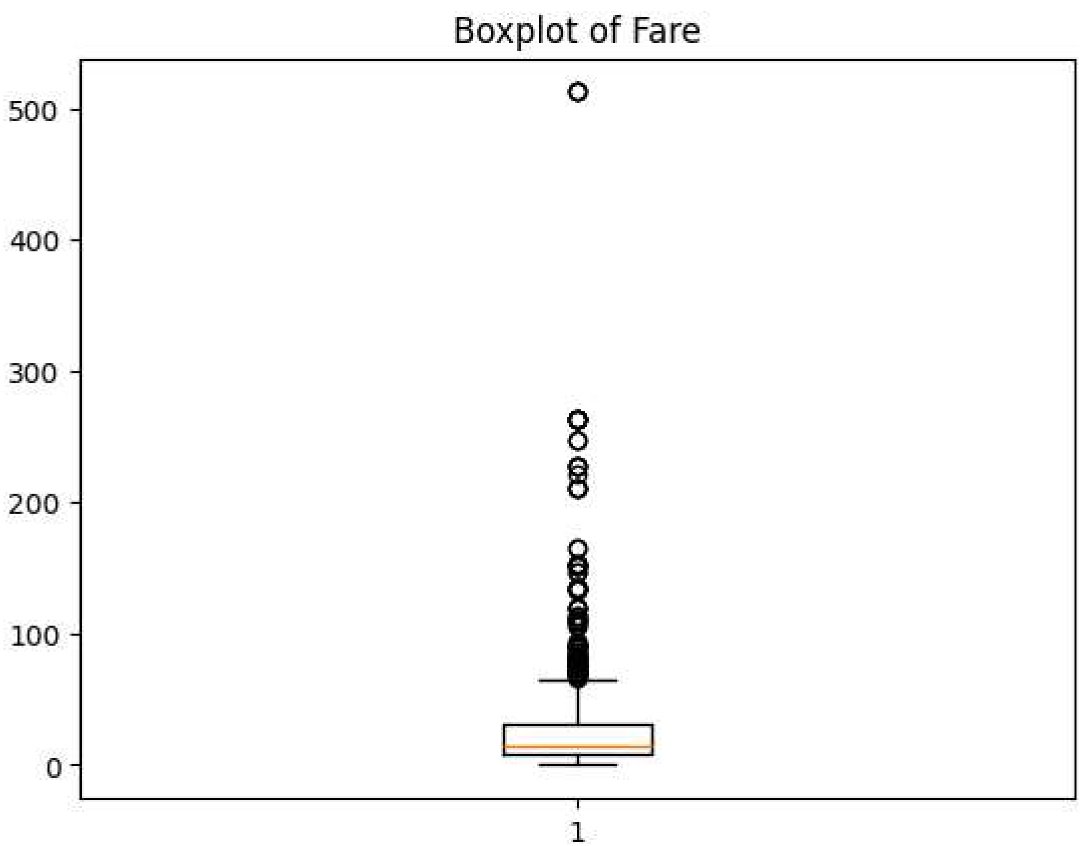
Out[45]:
```
survived       0
pclass         0
sex            0
age            0
sibsp          0
parch          0
fare           0
embarked       0
class          0
who            0
adult_male     0
embark_town    0
alive          0
alone          0
dtype: int64
```

In [47]:
```
df.dtypes
```

```
Out[47]: survived          int64
         pclass            int64
         sex              object
         age             float64
         sibsp             int64
         parch             int64
         fare            float64
         embarked         object
         class          category
         who              object
         adult_male         bool
         embark_town      object
         alive            object
         alone              bool
         dtype: object
```

```
In [48]: df['age']=df['age'].astype(int)
```

```
In [50]: pyplt.boxplot(df['fare'])
         pyplt.title('Boxplot of Fare')
         pyplt.show()
```




```
In [ ]: Q1 = df['fare'].quantile(0.25)
        Q3 = df['fare'].quantile(0.75)

        IQR = Q3-Q1

        lower = Q1-1.5*IQR
        upper = Q3+1.5*IQR
```

```
print('lower Limit:',lower)
print('upper limit:',upper)
```

```
lower Limit: -26.724
upper limit: 65.6344
```

Even though Fare cannot actually be negative, the IQR formula can still produce a negative lower bound.

In Titanic dataset:

No fare values are negative So practically only upper outliers matter

You can say in viva:

“The lower bound became negative due to IQR calculation, but since fare values cannot be negative, only values above the upper limit are treated as outliers.”

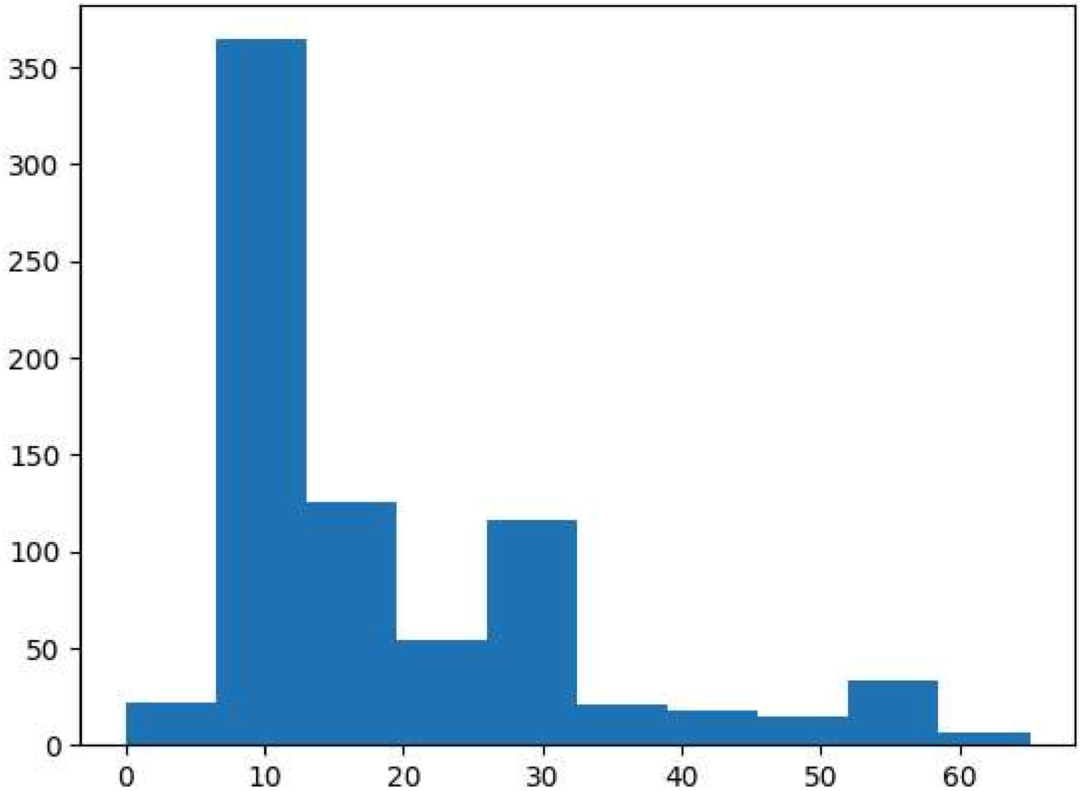
In [52]:
```
outliers=df[(df['fare']<lower)|(df['fare']>upper)]
print(outliers)
```

```
     survived  pclass     sex  age  sibsp  parch      fare embarked  class  \
1           1       1  female   38      1      0   71.2833        C  First   
27          0       1    male   19      3      2  263.0000        S  First   
31          1       1  female   29      1      0  146.5208        C  First   
34          0       1    male   28      1      0   82.1708        C  First   
52          1       1  female   49      1      0   76.7292        C  First   
..        ...     ...     ...  ...    ...    ...       ...      ...    ...   
846         0       3    male   29      8      2   69.5500        S  Third   
849         1       1  female   29      1      0   89.1042        C  First   
856         1       1  female   45      1      1  164.8667        S  First   
863         0       3  female   29      8      2   69.5500        S  Third   
879         1       1  female   56      0      1   83.1583        C  First   

       who  adult_male  embark_town alive  alone  
1    woman       False    Cherbourg   yes  False  
27     man        True  Southampton    no  False  
31   woman       False    Cherbourg   yes  False  
34     man        True    Cherbourg    no  False  
52   woman       False    Cherbourg   yes  False  
..     ...         ...          ...   ...    ...  
846    man        True  Southampton    no  False  
849  woman       False    Cherbourg   yes  False  
856  woman       False  Southampton   yes  False  
863  woman       False  Southampton    no  False  
879  woman       False    Cherbourg   yes  False  

[116 rows x 14 columns]
```

In [53]:
```
df=df[(df['fare']>=lower) & (df['fare']<=upper)]
```

In [56]:
```
pyplt.hist(df['fare'])
```

Out[56]:
```
(array([ 22., 364., 125.,  54., 116.,  21.,  18.,  15.,  33.,   7.]),
 array([ 0. ,  6.5, 13. , 19.5, 26. , 32.5, 39. , 45.5, 52. , 58.5, 65. ]),
 <BarContainer object of 10 artists>)
```



Observation

Fare is positively skewed.

Most values small, few very large.

In [57]:
```
df['fare_log']=np.log(df['fare']+1)
```

Explanation np.log()

Applies logarithm.

Why +1?

Avoids log(0) error.

Benefits

1. Reduces skewness
2. Makes distribution more normal
3. Compresses large values

In [58]:
```
df.head()
```

Out[58]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_m
0	0	3	male	22	1	0	7.2500	S	Third	man	T
2	1	3	female	26	0	0	7.9250	S	Third	woman	Fa
3	1	1	female	35	1	0	53.1000	S	First	woman	Fa
4	0	3	male	35	0	0	8.0500	S	Third	man	T
5	0	3	male	29	0	0	8.4583	Q	Third	man	T

Missing Values

Missing values:

NaN values Empty fields

Handling methods:

Mean Median Mode Remove rows/columns Outliers

Outliers:

Extreme abnormal values

Detection methods:

Boxplot IQR Z-score

Handling methods:

Removal Capping Transformation Data Transformation

Transforms variable distribution.

Used for:

Reducing skewness Better visualization Better ML performance Important Viva Questions Basic What is data wrangling?

Cleaning and transforming data.

What are outliers?

Extreme abnormal values.

Why handle outliers?

Improve accuracy.

What is skewness?

Lack of symmetry in distribution.

Intermediate Why log transformation used?

Reduces positive skewness.

Why add 1 before log?

To avoid log(0).

Difference between normalization and transformation? Normalization Transformation Scale change Distribution change External May Ask These Changes

1. Use Median Instead of Mean df['Age'] = df['Age'].fillna(df['Age'].median())
2. Detect Outliers Using Z-Score from scipy import stats

z = np.abs(stats.zscore(df['Fare'])) print(z) 3. Remove Duplicates df.drop_duplicates() 4. Standardize Data df['Fare_std'] = (df['Fare'] - df['Fare'].mean()) / df['Fare'].std()

Complete Theory of Outliers for Viva

1. What are Outliers?

Outliers are data points that are:

Extremely high Extremely low Different from most observations

They lie far away from normal data distribution.

Example

Suppose ages are:

18, 20, 22, 21, 19, 200

Here:

200

is an outlier.

2. Why Do Outliers Occur?

Outliers may occur because of:

Cause Example Data entry errors Age = 500 Measurement errors Wrong sensor reading Natural variation Billionaire income Experimental errors Faulty instrument 3. Types of Outliers A. Global Outliers

Very different from all other data.

Example:

Income = 1 crore

when most incomes are 20k–50k.

B. Contextual Outliers

Outlier only in specific context.

Example:

Temperature 40°C in winter C. Collective Outliers

A group of unusual observations together.

Used in:

Fraud detection Network intrusion detection 4. Why Outliers are Problematic?

Outliers can:

Distort mean Affect standard deviation Reduce model accuracy Mislead analysis Example

Data:

10, 12, 14, 16, 500

Mean becomes very large because of:

500 5. Effect of Outliers on Statistics Statistical Measure Effect Mean Highly affected Median Less affected Standard deviation Highly affected Viva Question Why median is preferred when outliers exist?

Because median is robust and less affected by extreme values.

6. How to Detect Outliers?

Common methods:

Method Used For Boxplot Visualization IQR Method Skewed data Z-score Normal distribution Scatter plot Visual detection 7. Boxplot Theory

Boxplot is graphical representation of:

Median Quartiles Outliers Parts of Boxplot Part Meaning Box Middle 50% data Line inside box Median Whiskers Normal range Dots outside whiskers Outliers Viva Question What does box represent?

Interquartile range (Q1 to Q3).

8. Quartiles Theory

Quartiles divide data into 4 equal parts.

Quartile Meaning Q1 25% Q2 Median Q3 75% 9. IQR Theory

IQR = Interquartile Range

Formula:

$IQR = Q3 - Q1$

Measures spread of middle 50% data.

Why IQR Used?

Advantages:

Resistant to outliers Works well for skewed distributions Simple and effective 10. IQR Outlier Detection Formula

Lower limit:

$Q1 - 1.5 \times IQR$

Upper limit:

$Q3 + 1.5 \times IQR$

Outlier Rule

If:

value < lower limit

OR

value > upper limit

then it is an outlier.

11. Why 1.5 Used in IQR?

1.5 is standard statistical threshold.

It balances:

Sensitivity False detection Viva Question Why 1.5 × IQR?

It is statistically accepted range for detecting moderate outliers.

12. Z-Score Method

Measures how far value is from mean.

Formula:

$z = x - \mu / \sigma$

Where:

x = data value μ = mean σ = standard deviation Z-Score Rule

If:

$|z| > 3$

then value is considered outlier.

Viva Question Difference between IQR and Z-score? IQR Z-score Uses quartiles Uses mean/std Good for skewed data Good for normal data Robust Sensitive to outliers 13. Handling Outliers A. Removing Outliers

Delete abnormal values.

Used when:

Errors exist Few outliers present B. Capping/Winsorization

Replace extreme values with boundary values.

Example:

1000 → 95th percentile value C. Transformation

Apply:

Log transformation Square root transformation

Used to reduce skewness.

D. Keep Outliers

Sometimes outliers contain important information.

Example:

Fraud transactions Disease detection Viva Question Should we always remove outliers?

No.

Sometimes outliers are meaningful and important.

14. Log Transformation and Outliers

Formula:

y=log(x+1)

Benefits:

Compresses large values Reduces skewness Reduces effect of outliers 15. Skewness Theory

Skewness measures asymmetry of distribution.

Positive Skew

Tail on right side.

Example:

Income distribution Negative Skew

Tail on left side.

Viva Question Which skewness does log transformation reduce?

Positive/right skewness.

16. Difference Between Noise and Outliers Noise Outlier Random error Extreme value Usually meaningless May be meaningful
17. Real-Life Applications of Outlier Detection

Used in:

Fraud detection Credit card analysis Healthcare Cybersecurity Stock market analysis 18. Important Outlier Viva Questions Basic Questions What is an outlier?

Extreme abnormal data point.

Why detect outliers?

To improve analysis accuracy.

Which graph is commonly used?

Boxplot.

Which statistical method used?

IQR and Z-score.

Intermediate Questions Why mean affected by outliers?

Because mean uses all values.

Why median resistant to outliers?

Median depends on position not magnitude.

What is IQR?

Difference between Q3 and Q1.

Why boxplot preferred?

Easy visual detection.

What are whiskers in boxplot?

Range of normal values.

Advanced Viva Questions Can outliers be useful?

Yes:

Fraud detection Rare disease analysis Anomaly detection Which method is better: IQR or Z-score?

Depends on distribution:

IQR → skewed data Z-score → normal distribution Why lower limit may become negative?

Because formula:

Q1 - 1.5 × IQR

can mathematically produce negative values.

External May Ask Code Changes Find Outliers in Age Q1 = df['Age'].quantile(0.25) Q3 = df['Age'].quantile(0.75)

IQR = Q3 - Q1

lower = Q1 - 1.5*IQR upper = Q3 + 1.5*IQR

df[(df['Age'] < lower) | (df['Age'] > upper)] Use Z-score from scipy import stats

z = np.abs(stats.zscore(df['Fare']))

df[z > 3] Plot Histogram plt.hist(df['Fare']) plt.show() Short Viva Explanation (Best Answer)

“Outliers are extreme data points significantly different from the rest of the dataset. They may occur due to errors or natural variations. Outliers can affect statistical measures like mean and standard deviation and reduce machine learning accuracy. Common methods to detect outliers are boxplot, IQR method, and Z-score method. In this practical, we used the

IQR method where values outside the lower and upper limits are considered outliers. We handled them by removing extreme values and applying transformations.”